

SQL Quiz



SQL Quiz

Database Fundamentals



The wheel picks the student — **one student, one question!**

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Question 1

SQL

What does the `WHERE` **clause do in** `SELECT * FROM users`
`WHERE age > 18; ?`

1 Sorts the results by age

2 Groups rows that share the same age

3 Selects only the age column

4 Filters rows so only those matching the condition are returned



Question 1

SQL

What does the WHERE clause do in `SELECT * FROM users WHERE age > 18; ?`

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2 Groups rows that share the same age

3 Selects only the age column

4 Filters rows so only those matching the condition are returned



✓ Answer 4 — WHERE filters rows in the source table before they're returned — only rows where age > 18 make it into the result set



What does an `INNER JOIN` return?

- 1 All rows from both tables, matching or not
- 2 Only rows that have matching values in both joined tables
- 3 Only rows from the left table
- 4 A random sample of rows from both tables



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2 Only rows that have matching values in both joined tables ✓

3 Only rows from the left table

4 A random sample of rows from both tables

✓ Answer 2 — `INNER JOIN` returns only the rows where the join condition matches in both tables — non-matching rows are excluded



What's the difference between WHERE and HAVING ?

1 They do exactly the same thing

2 WHERE filters after grouping,
HAVING filters before grouping

3 WHERE filters rows before
grouping, HAVING filters groups
after aggregation

4 HAVING can only be used with
DELETE statements



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✓ Answer 3 — WHERE filters individual rows before GROUP BY runs;
HAVING filters the aggregated groups afterward (e.g. HAVING
COUNT(*) > 5)



What is a PRIMARY KEY for?

- 1 Uniquely identifies each row in a table and cannot be NULL
- 2 Links a table to another table
- 3 Speeds up INSERT statements only
- 4 Encrypts sensitive columns



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1 Uniquely identifies each row in a table and cannot be NULL ☒

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3 Speeds up INSERT statements only

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☒ Answer 1 — A PRIMARY KEY uniquely identifies every row in a table — it must be unique and can never be NULL



What does a FOREIGN KEY enforce?

- 1 That a column must always be unique
- 2 That a table can only have one row
- 3 That queries run faster
- 4 That a column's values must match an existing value in another table (referential integrity)



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Why does `column = NULL` never match any rows?

1

Because NULL columns are automatically excluded from all queries

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Because NULL means "unknown", and comparing anything to an unknown value is never true — use `IS NULL` instead

3

Because NULL is stored as the number 0

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Because `=` only works on numeric columns



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What does `ORDER BY price DESC LIMIT 5` **do?**

1 Deletes the 5 cheapest rows

2 Groups rows into 5 buckets by price

3 Sorts rows by price from highest to lowest and returns only the top 5

4 Updates the price of 5 rows



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4 Updates the price of 5 rows

✓ Answer 3 — `ORDER BY price DESC` sorts descending by price, and `LIMIT 5` keeps only the first 5 rows of that sorted result



What's the key difference between DELETE and TRUNCATE ?

1

DELETE can remove specific rows with a WHERE clause and is logged row-by-row; TRUNCATE removes all rows at once with minimal logging

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TRUNCATE can filter rows with WHERE, DELETE cannot

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They are exactly the same, just different keywords

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DELETE removes the table structure, TRUNCATE only removes data



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They are exactly the same, just different keywords

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Answer 1 — DELETE supports WHERE and logs each row removed (so it can be rolled back); TRUNCATE wipes the whole table at once



What does `SELECT DISTINCT country FROM customers` **return?**

- 1 Every row from the customers table
- 2 Only customers whose country is NULL
- 3 The count of customers per country
- 4 Each unique country value, with duplicates removed



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2 Only customers whose country is NULL

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4 Each unique country value, with duplicates removed



✓ **Answer 4** — `DISTINCT` removes duplicate values from the result set — you get each unique country only once



What does `SELECT department, COUNT(*) FROM employees GROUP BY department;` **return?**

- 1 The total number of employees in the whole company
- 2 One row per department, with the count of employees in each
- 3 Every employee row, unsorted
- 4 An error, because COUNT(*) can't be used with GROUP BY



Question 10

SQL

What does `SELECT department, COUNT(*) FROM employees GROUP BY department;` **return?**

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2 One row per department, with the count of employees in each ✓

3 Every employee row, unsorted

4 An error, because COUNT(*) can't be used with GROUP BY

✓ Answer 2 — GROUP BY collapses rows that share the same department value into one group per department, and COUNT(*) counts the rows in each group





Well Done!

That's the end of the SQL quiz.

SELECT, JOIN, GROUP BY/HAVING, keys, NULL, ORDER BY/LIMIT,
DELETE/TRUNCATE, DISTINCT: ✓